



香港中文大學(深圳)
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Introduction to Data Science

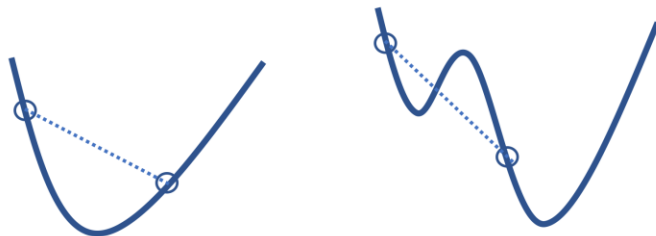
Lecture 20 Optimization V: Practice Questions (Continued)

- The goal of today's lecture is to guide you through a few questions and reinforce your understanding of
 - 1) how to prove a convex function
 - 2) how to formulate an optimization problem.

Review: How to Check a Function is Convex?

You can consider using:

- **Visualization** – If the function is univariate, we can use a geometric approach to check convexity.



- **Definition-based proof** – If $\text{dom}(f)$ is a convex set, choose
 $\forall x_1, x_2 \in \text{dom}(f), \forall \theta \in [0,1]$
$$f(\theta x_1 + (1 - \theta)x_2) \leq \theta f(x_1) + (1 - \theta)f(x_2).$$

Review: How to Check a Function is Convex?

You can consider using:

- **Second-Order Condition** – Suppose $f: R^n \rightarrow R$ is twice continuously differentiable (smooth). Then f is convex if and only if for any $\mathbf{x}$, for every direction $\mathbf{e}$, it holds that

$$\frac{d^2 f(\mathbf{x} + t\mathbf{e})}{dt^2} \Big|_{t=0} \geq 0$$

Review: How to Check a Function is Convex?

You can consider using:

- **Operations that Preserve Convexity**

- If f is convex, then f/c is convex for any $c > 0$
- If f_1, f_2 are convex, then $f_1 + f_2$ are convex
- **Affine Transformation:** If f is convex, then $g(\mathbf{x}) = f(a\mathbf{x} + \mathbf{b}) - c$ is also convex
- **Pointwise Maximum:** If $f_1, \dots, f_m$ are convex, then the function

$$f(\mathbf{x}) = \max\{f_1(\mathbf{x}), \dots, f_m(\mathbf{x})\}$$

is also convex.

- **Non-negative Weighted Sums:** If $f_1, f_2, \dots, f_n$ are convex functions, and $w_1, w_2, \dots, w_n \geq 0$, then the function

$$f(x) = w_1 f_1(x) + w_2 f_2(x) + \dots + w_n f_n(x)$$

is convex.

Lecture 20: Optimization VI: Practice Questions (Continued)

Outline

Exercise 1 (Taken from Our Assignment)

Exercise 2 (Taken from Our Assignment)

Exercise 3 (Taken from Last Year's Final Exam)

Exercise 4 (Taken from Last Year's Final Exam)

Exercise 5 (Taken from Last Year's Final Exam)

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Exercise 1:

- **Question:** Prove the following: if f is convex in (x, y) , and C is a convex nonempty set, then the function

$$g(x) = \inf_{y \in C} f(x, y)$$

is convex.

This question is taken from Assignment 3.

Solution to Exercise 1

► **Solution:** Pick two points: x_1 and x_2 . Let

$$y_1 = \arg \min_{y \in C} f(x_1, y), \quad y_2 = \arg \min_{y \in C} f(x_2, y)$$

so that

$$g(x_1) = f(x_1, y_1), \quad g(x_2) = f(x_2, y_2)$$

Because C is convex set in y , the convex combination

$$y_\lambda = \lambda y_1 + (1 - \lambda) y_2 \in C$$

Solution to Exercise 1

- **Solution:** Now define $x_\lambda = \lambda x_1 + (1 - \lambda)x_2$. Using convexity of $f(x, y)$, we get:

$$\begin{aligned} f(x_\lambda, y_\lambda) &\leq \lambda f(x_1, y_1) + (1 - \lambda)f(x_2, y_2) \\ &= \lambda g(x_1) + (1 - \lambda)g(x_2) \end{aligned}$$

Now, since $g(x_\lambda) = \min_{y \in C} f(x_\lambda, y)$, we have:

$$g(x_\lambda) \leq f(x_\lambda, y_\lambda) \leq \lambda g(x_1) + (1 - \lambda)g(x_2).$$

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Exercise 5 (Taken from Last Year's Final Exam)

- **Question:** Show the following function is convex: $f(x) = \sum_{i=1}^r |x|_{[i]}$ on $\mathbb{R}^n$, where $|x|$ denotes the vector with $|x|_i = |x_i|$ (i.e., $|x|$ is the absolute value of x , componentwise), and $|x|_{[i]}$ is the i -th largest component of $|x|$. In other words, $|x|_{[1]}, |x|_{[2]}, \dots, |x|_{[n]}$ are the absolute values of the components of x , sorted in non-increasing order.

This question is taken from Assignment 3.

Solution to Exercise 2

- Solution: Write f as

$$f(x) = \sum_{i=1}^r |x|_{[i]} = \max_{1 \leq i_1 < i_2 < \dots < i_r \leq n} |x_{i_1}| + |x_{i_2}| + \dots + |x_{i_r}|$$

which is the pointwise maximum of $\frac{n!}{r!(n-r)!}$ convex function.

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Exercise 3: Convex Function

► (10 points) Convex Functions

- (a) (5 points) Given a vector $\mathbf{x} \in \mathbb{R}^n$, let $|\mathbf{x}|$ denote the vector with $|\mathbf{x}|_i = |x_i|$ (i.e., $|\mathbf{x}|$ is the absolute value of $\mathbf{x}$, componentwise). Let $|\mathbf{x}|_{[i]}$ denote the i -th largest component of $|\mathbf{x}|$. In other words, $|\mathbf{x}|_{[1]}, |\mathbf{x}|_{[2]}, \dots, |\mathbf{x}|_{[n]}$ are the absolute values of the components of $\mathbf{x}$, sorted in non-increasing order.
Show that $f(\mathbf{x}) = \sum_{i=1}^r \alpha_i |\mathbf{x}|_{[i]}$ is a convex function of $\mathbf{x}$, where $\alpha_i = 2n - i$ and $r \leq n$ is a positive integer. Explain the reasons.
- (b) (5 points) For what range of a will function $f(x) = ax^3 + 6 \ln x$ be a real-valued convex function within interval (a^2, ∞) ? Explain the reasons.

This question is taken from last year's final exam.

Solution to Exercise 3

- (a) We can apply the previous question's result to this question: We want to show that:

$$f(x) = \sum_{i=1}^r \alpha_i |x|_{[i]}$$

is a convex function, where $\alpha_i = 2n - i$ (positive constants) and $r \leq n$.

Note here α_i is non-decreasing with i .

When $r = 1$, we have shown in Example 2 that $|x|_{[1]}$ is convex, and hence $\alpha_1 |x|_{[1]}$ is also convex because scaling by a positive constant preserves the convexity.

Solution to Exercise 3

- When $r = 2$, observe that $\alpha_1|x|_{[1]} + \alpha_2|x|_{[2]} = (\alpha_1 - \alpha_2)|x|_{[1]} + \alpha_2(|x|_{[1]} + |x|_{[2]})$. From Example 2 we know that both the function $|x|_{[1]}$ and the function $|x|_{[1]} + |x|_{[2]}$ are convex, thus their sum is also convex.

More generally, for a generic r , we have

$f(x) = \cdots |x|_{[1]} + \cdots (|x|_{[1]} + |x|_{[2]}) + \cdots + (\alpha_{r-1} - \alpha_r)(|x|_{[1]} + |x|_{[2]} + \cdots + |x|_{[r-1]}) + \alpha_r(|x|_{[1]} + |x|_{[2]} + \cdots + |x|_{[r]})$. Each summand is convex thanks to Example 2. Hence their sum is also convex

Solution to Exercise 3

- (b) We are tasked with finding the range of a for which the function

$$f(x) = ax^3 + 6 \ln x$$

is convex within the interval (a^2, ∞) . We can derive that

$$f''(x) = 6ax - \frac{6}{x^2}$$

For the function to be convex, we require $f''(x) \geq 0$. Therefore, we need to solve the inequality:

$$6ax - \frac{6}{x^2} \geq 0$$

Solution to Exercise 3

- ▶ Divide both sides by 6 to simplify:

$$ax - \frac{1}{x^2} \geq 0$$

Multiply through by x^2 (which is positive for $x > 0$):

$$ax^3 - 1 \geq 0$$

This simplifies to:

$$ax^3 \geq 1$$

Thus, for convexity, we need:

$$a \geq \frac{1}{x^3}$$

Solution to Exercise 3

- We are given that the function is considered on the interval (a^2, ∞) , meaning $x \in (a^2, \infty)$. For convexity, the function $ax^3 \geq 1$ must hold for $x \in (a^2, \infty)$. The smallest value of x in this interval is $x = a^2$. Thus, we require:

$$a(a^2)^3 \geq 1$$

Simplifying:

$$aa^6 \geq 1 \quad \Rightarrow \quad a^7 \geq 1$$

This implies:

$$a \geq 1$$

- Next, let's review

- 1) how to **formulate** a newsvendor problem

- 2) how to **find the optimal solution** for the newsvendor problem

Review: Newsvendor Problem

- A newsvendor must decide how many newspapers to purchase each morning to maximize expected profit.
 - Demand D is random, with probability density function $f(\cdot)$ and cumulative distribution function $F(\cdot)$.
 - Costs & Revenues
 - Purchase cost per newspaper: c
 - Selling price per newspaper: p
 - Salvage value: 0 (unsold newspapers have no residual value)
 - Decision variable: Order quantity q



Review: Newsvendor Problem

- Objective

- Sales: $\min(q, D)$
- Expected revenue: $E_D[p \cdot \min(q, D)]$
- Ordering cost: cq
- Expected profit

$$\pi(q) = E_D[p \min(q, D)] - cq$$

- *Exercise: Show π is concave, or equivalently, $-\pi$ is convex.*
 - Hint: using operations that preserve convexity



Review: Newsvendor Problem

Proof. (of convexity of $-\pi(q) = -E_D[p \min(q, D)] + cq$)

1. The function $g(q) = -\min(q, D)$ is convex in q
2. The function $h(g(q)) = pg(q)$ is convex in q
3. The expectation $-E_D[p \min(q, D)]$ is convex in q
4. The objective $-\pi(q)$ is convex

Review: Newsvendor Problem

Proof. (of convexity of $-\pi(q) = -E_D[p \min(q, D)] + cq$)

1. The function $g(q) = -\min(q, D)$ is convex in q
 - Because $g(q) = -\min(q, D) = \max(-q, -D)$ is the maximum of linear functions of q
2. The function $h(g(q)) = pg(q)$ is convex in q
 - Because scaling by a positive constant $p > 0$ preserves convexity
3. The expectation $E_D[p \min(q, D)]$ is convex in q
 - Because weighted sum of convex functions is still convex
4. The objective $-\pi(q)$ is convex
 - Because cq is linear (thus convex), the sum preserves convexity

Review: Newsvendor Problem: Finding the Optimal Order Quantity

- Express the objective as an integral

$$\pi(q) = p \int_0^q x f(x) dx + pq(1 - F(q)) - cq.$$

- To maximize $\pi(q)$ take the derivative with respect to q :

$$\frac{d}{dq}\pi(q) = p \cdot qf(q) + p(1 - F(q)) - pqf(q) - c$$

- Simplifies to:
$$\frac{d}{dq}\pi(q) = p(1 - F(q)) - c$$

Review: Newsvendor Problem: Finding the Optimal Order Quantity

- Setting $\frac{\partial}{\partial q} \pi(q) = 0$:

$$F(q^*) = \frac{p - c}{p}.$$

- Solving for q^* :

$$q^* = F^{-1} \left(\frac{p - c}{p} \right)$$

 The optimal order quantity is directly linked to demand uncertainty and price-cost ratio.

 This reinforces why convex optimization is a powerful tool in real-world decision-making! 

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Exercise 4

- **Question:** An airline company would like to calculate the **demand** when it sets the **price** of the flight from Beijing to Shanghai as p . Suppose an arriving potential customer would purchase the ticket if and only if his willingness-to-pay (WTP) is higher than the posted price p . Also assume that a potential customer's WTP X is random, following a distribution with a continuous cumulative distribution function

$$F(x) = \mathbb{P}(X \leq x)$$

It is known that $F(x) = 0$ if $x \leq 1$.

This question is taken from last year's final exam

Exercise 4

- ▶ (a) (2 points) For the random experiment of an arriving potential customer buying the ticket or not at price $p \geq 1$, describe the sample space and the probability associated with each outcome.

Exercise 4

- ▶ (b) (2 points) Suppose there is a total of n arriving potential customers, and each customer's purchase decision is independent of each other. Consider the random variable D denoting the total number of sales from n arriving customers under posted price $p \geq 1$ (assume that the total number of flight tickets to sell is larger than n). What is the distribution of D ? Give its mean and variance.

Exercise 4

- ▶ (c) (3 points) Consider the mean of D in the previous question. Obviously, it depends on the posted price p . We denote the **expected demand** as $d(p)$. Suppose for each ticket sold, the airline company incurs a cost of $c \geq 0$ to prepare the flight service. If the airline company wants to set its posted price $p \geq 1$ to maximize the expected profit, write down the formulation of this optimization problem.
 - ▶ i. What's the decision variable?
 - ▶ ii. What's the objective function?
 - ▶ iii. What's the constraint?

Exercise 4

- ▶ (d) (3 points) The profit of the previous part depends on the customers' WTP distribution F . Suppose F is twice continuously differentiable. Give the second order condition under which the profit-maximization problem's objective is concave in $p \in [1, \infty)$.

Exercise 4

- ▶ (e) The company would like to estimate the relationship between mean demand $d(p)$ and price $p \geq 1$ using the constant elasticity demand function $d(p) = ap^b$, where $a > 0$ and $b < 0$ are parameters to be determined.
 - ▶ i. (4 points) Given that $d(p) = ap^b$ for $p \geq 1$, calculate the value of n and the customer's WTP distribution F .
 - ▶ ii. (4 points) If the price is set as p , the expected consumer surplus of a representative consumer is defined as $\mathbb{E}[\max\{X - p, 0\}]$. Calculate the expected consumer surplus for one representative consumer when $d(p) = ap^b$ and $b < -1$.

Exercise 4

- ▶ (e)
 - ▶ iii. (4 points) Given past price and demand data $(D_1, p_1), (D_2, p_2), \dots, (D_m, p_m)$ under m different price experiments (for example, under price p_1 , the demand is D_1), the company would like to estimate the values of the parameters a and b using MLE. More specifically, it is assumed that the actual demand under posted price p is normally distributed with mean ap^b and known variance σ^2 (σ is small enough so that the chance of demand being negative can be neglected). Write down the loglikelihood function to estimate the parameters a and b . (You don't need to solve it.)

Solution to Exercise 4

- **Solution:** (a) The sample space is

$$\{\text{buy, not buy}\}$$

The probabilities are

$$P(\text{not buy}) = P(X \leq p) = F(p)$$

and

$$P(\text{buy}) = P(X > p) = 1 - F(p)$$

(b) D is Binomial($n, 1 - F(p)$), with mean $n(1 - F(p))$ and variance $n(1 - F(p))F(p)$.

Solution to Exercise 4

- **Solution:** (c) The decision variable is the price p . The objective function is the expected profit

$$n(1 - F(p))(p - c)$$

The constraint is $p \geq 1$.

(d) The second order derivative of the objective function is $-nF''(p)(p - c) - 2nF'(p)$. To ensure concavity of the objective function, we need $-nF''(p)(p - c) - 2nF'(p) \leq 0$.

Solution to Exercise 4

- **Solution:** (e) i. From part b we know that $d(p) = n(1 - F(p))$, hence $n(1 - F(p)) = ap^b$, i.e., $F(p) = 1 - \frac{ap^b}{n}$. Since $F(1) = 0$, we have $1 - a/n = 0$, i.e., $n = a$. Therefore $F(p) = 1 - p^n$.
- ii. The expected consumer surplus of one representative consumer is

$$\begin{aligned}\mathbb{E}[\max\{X - p, 0\}] &= \int_p^\infty (x - p)F'(x)dx \\ &= \int_p^\infty -(x - p)bx^{b-1}dx \\ &= -\frac{1}{b+1}p^{b+1}\end{aligned}$$

- iii. Under price p_i , the demand distribution's pdf is

$$f(D_i | p_i) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{D_i - ap_i^b}{\sigma}\right)^2}$$

Solution to Exercise 4

- **Solution (Continued):** Therefore, the likelihood function is

$$\begin{aligned} L(a, b) &= \prod_{i=1}^m f(D_i | p_i) \\ &= \frac{1}{(\sigma\sqrt{2\pi})^m} e^{-\frac{1}{2} \sum_{i=1}^m \left(\frac{D_i - ap_i^b}{\sigma} \right)^2} \end{aligned}$$

and the log-likelihood function is

$$l(a, b) = -m \log(\sigma\sqrt{2\pi}) - \frac{1}{2} \sum_{i=1}^m \left(\frac{D_i - ap_i^b}{\sigma} \right)^2$$

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Exercise 5:

► **Question:** (16 points) **The Newsvendor's Problem: MLE + Convex Optimization**

A newspaper vendor faces the task of determining the optimal quantity of newspapers to stock, given uncertain demand D . The cost structure for ordering newspapers is as follows: the first twelve copies incur a cost of \$8 per copy, while any additional copies beyond twelve incur a cost of \$5 per copy. Each sold copy yields a revenue of \$10 for the vendor, while each unsold copy has a salvage value of \$2. Remark: for ease of analysis, we allow the vendor to order a non-integer number of copies.

Exercise 5:

► Question:

- (a) (4 points) Given that the demand D follows a uniform distribution with a support of $[\theta, 2\theta]$, where θ is unknown, the objective is to estimate the maximum likelihood estimator (MLE) of θ . Five samples of D are provided: $\{12, 15, 13, 20, 14\}$. Calculate the MLE of θ . Explain the result in detail.
- (b) (8 points) Assuming that D follows a uniform distribution with a support of $[\theta, 2\theta]$, where θ is the MLE obtained in (a), calculate the vendor's optimal order decision. Explain the result in detail.
- (c) (4 points) For (b), what's the vendor's optimal order decision, assuming that the vendor can only order an integer number of copies? Explain the result in detail.

Solutin to Exercise 5

► **Solution:**

- (a) The likelihood is $\frac{1}{\theta^5}$ if $\theta \leq 12$ and $2\theta \geq 20$; otherwise, the likelihood is 0. To maximize the likelihood, we need to set $\theta = 10$.

Solutin to Exercise 5

► Solution:

- (b) The demand $D \sim \mathcal{U}[10, 20]$. As the minimum of concave functions is concave and the non-negative weighted sum of concave functions is concave, $\mathbb{E}[\min(Q, D)]$ is concave in Q . i. If $Q \leq 12$, the revenue is $10\mathbb{E}[\min(Q, D)]$, the salvage value is $2(Q - \mathbb{E}[\min(Q, D)])$, the cost is $8Q$. Thus, the profit is as follows:

$$8\mathbb{E}[\min(Q, D)] - 6Q$$

which is concave. Thus, if the optimal solution satisfies $Q^* < 12$, by the first-order optimality condition, the optimal decision shall be

$$\frac{20 - Q^*}{10} = \frac{6}{8}$$

namely $Q^* = 12.5$. However, $12.5 \geq 12$. Thus, the optimal decision shall be $Q^* = 12$.

Solutin to Exercise 5

► Solution:

- ii. If $Q \geq 12$, the revenue is $10\mathbb{E}[\min(Q, D)]$, the salvage value is $2(Q - \mathbb{E}[\min(Q, D)])$, the cost is $5Q + 36$. Thus, the profit is as follows:

$$8\mathbb{E}[\min(Q, D)] - 3Q - 36$$

which is concave. Thus, if the optimal solution satisfies $Q^* > 12$, by the first-order optimality condition, the optimal decision shall be

$$\frac{20 - Q^*}{10} = \frac{3}{8}$$

namely $Q = 16.25$. As $16.25 \geq 12$, the optimal decision shall be $Q^* = 16.25$. iii. As the cost is continuous in Q , the profit is continuous in Q . (i) implies that $\pi(Q) \leq \pi(12)$ for $Q \leq 12$ while (ii) implies that $\pi(65/4) \geq \pi(Q)$ for $Q \geq 12$. Thus the optimal Q shall be $65/4$.

Solutin to Exercise 5

► **Solution:**

- (c) The optimal order quantity shall be either 16 or 17. $\pi(16) = 148/5$ and $\pi(17) = 147/5$. Thus the optimal order decision is 16 copies.